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Q3 -0
Q2 -0
Q4 -0
Q5 -0
Q1 -0

Homework Assignment 4

Show all your computations and justify your answers.

1. (20 points) Show that the curve $f(x) = 2\ln x + x^2$ has no tangent line parallel to $y = 2x$.

Prove:

$f(x)=2\ln x+x^2$, the domain is $x>0$

$f'(x)=\frac{2}{x}+2x \ (x>0)$

Assume that $f(x)$ has tangent line parallel to $y=2x$

that is: $f'(x)=2 \Rightarrow \frac{2}{x}+2x=2$

$$2+2x^2=2x$$

$$x^2-x+1=0$$

$$\Delta=(-1)^2-4\times1\times1=-3<0$$

That means there are no real number can satisfy $x^2-x+1=0$

In conclusion, $f(x)=2\ln x+x^2$ has no tangent line parallel to $y=2x$

2. (20 points) Consider the function $f(x) = e^{2x}(\cos 3x - \sin 3x)$. Prove that $f''(x) - 4f'(x) + 13f(x) = 0$ for any x in $\mathbb{R}$.

Prove:

$f(x) = e^{2x}(\cos 3x - \sin 3x)$

$f'(x) = (e^{2x})'(\cos 3x - \sin 3x) + e^{2x}(\cos 3x - \sin 3x)'$

$= 2e^{2x}(\cos 3x - \sin 3x) + e^{2x}(-3\sin 3x - 3\cos 3x)$

$= -5e^{2x}\sin 3x - e^{2x}\cos 3x$

$= -e^{2x}(5\sin 3x + \cos 3x)$

$f''(x) = (-e^{2x})'(5\sin 3x + \cos 3x) + (-e^{2x})(5\sin 3x + \cos 3x)'$

$= -2e^{2x}(5\sin 3x + \cos 3x) - e^{2x}(15\cos 3x - 3\sin 3x)$

$= -10e^{2x}\sin 3x - 2e^{2x}\cos 3x - 15e^{2x}\cos 3x + 3e^{2x}\sin 3x$

$= -7e^{2x}\sin 3x - 17e^{2x}\cos 3x$

$= -e^{2x}(7\sin 3x + 17\cos 3x)$

so $f''(x) - 4f'(x) + 13f(x)$ is:

$-e^{2x}(7\sin 3x + 17\cos 3x) - 4\cdot(-e^{2x})(5\sin 3x + \cos 3x) + 13e^{2x}(\cos 3x - \sin 3x)$

$= (-7 + 20 - 13)e^{2x}\sin 3x + (-17 + 4 + 13)e^{2x}\cos 3x$

$= 0 + 0$

$= 0$

3. (20 points) Let f be a differentiable function on $\mathbb{R}$ such that $f\left(\frac{\pi}{2}\right) = -2\pi$, $f'\left(\frac{\pi}{2}\right) = 2$ and $f'(\pi(2-\pi)) = -1$. For $g(x) = f((x\sin x - 1)f(x))$, find $g'\left(\frac{\pi}{2}\right)$.

$$g(x) = f((x\sin x - 1)f(x))$$

$$g'(x) = \left[(x\sin x - 1)f(x)\right]' \cdot f'((x\sin x - 1)f(x))$$

$$= \left\{\left[(x\sin x)'f(x) + (x\sin x)f'(x)\right] - f'(x)\right\} \cdot f'((x\sin x - 1)f(x))$$

$$= \left\{\left[(\sin x + x\cos x)f(x) + (x\sin x)f'(x)\right] - f'(x)\right\} \cdot f'((x\sin x - 1)f(x))$$

$$g'\left(\tfrac{\pi}{2}\right) = \left\{\left[\left(\sin\tfrac{\pi}{2} + \tfrac{\pi}{2}\cos\tfrac{\pi}{2}\right)f\left(\tfrac{\pi}{2}\right) + \left(\tfrac{\pi}{2}\sin\tfrac{\pi}{2}\right)f'\left(\tfrac{\pi}{2}\right)\right] - f'\left(\tfrac{\pi}{2}\right)\right\} \cdot f'\left(\left(\tfrac{\pi}{2}\sin\tfrac{\pi}{2} - 1\right)f\left(\tfrac{\pi}{2}\right)\right)$$

$$= \left\{\left[\left(1 + \tfrac{\pi}{2}\cdot 0\right)\cdot(-2\pi) + \left(\tfrac{\pi}{2}\cdot 1\right)\cdot 2\right] - 2\right\} \cdot f'\left(\left(\tfrac{\pi}{2}\cdot 1 - 1\right)\cdot(-2\pi)\right)$$

$$= \left[-2\pi + \pi - 2\right] \cdot f'(-\pi^2 + 2\pi)$$

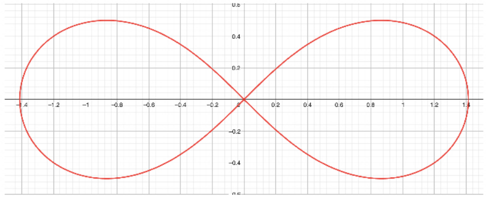
$$= (-\pi - 2) \cdot f'(\pi(2-\pi))$$

$$= (-\pi - 2) \cdot (-1)$$

$$= \pi + 2$$

✓

4. (20 points) Find all the horizontal tangent lines of the curve $(x^2+y^2)^2 = 2x^2 - 2y^2$.



According to the equation, we can find the image is symmetrical about the origin.

That means to find horizontal tangent line, we can first focus on $y>0$.

Horizontal tangent line $\Rightarrow y'=0$

$$(x^2+y^2)^2 = 2x^2-2y^2$$

$$x^4+2x^2y^2+y^4 = 2x^2-2y^2$$

then we take the derivative of both sides

$$(x^4+2x^2y^2+y^4)' = (2x^2-2y^2)'$$

$$4x^3+2(2xy^2+x^2(y^2)') + 4y^3\cdot y' = 4x-2(y^2)'$$

$$4x^3+4xy^2+2x^2y\cdot y'+4y^3\cdot y' = 4x-4y\cdot y'$$

Assume $y'=0$, then $4x^3+4xy^2=4x$

$$4x(x^2+y^2-1)=0$$

Since $x=0$, $y=0$ isn't the horizontal tangent line

$x^2+y^2=1$ ①

then we take ① into equation

that is $1=2x^2-2y^2$ ②

According to ① ②

$$\begin{cases} x^2+y^2=1 \\ 2x^2-2y^2=1 \end{cases} \Rightarrow \begin{cases} x=\pm\frac{\sqrt{3}}{2} \\ y=\pm\frac{1}{2} \end{cases}$$

In conclusion, $y=\pm\frac{1}{2}$ are the horizontal tangent lines.

5. (20 points) Find the derivative of $f(x) = \frac{\log_2(x^2+2)}{2\sqrt{x}}$.

The derivative of $f(x)$ is:

$$f'(x) = \left(\frac{\log_2(x^2+2)}{2^{\sqrt{x}}}\right)'$$

$$= \frac{[\log_2(x^2+2)]' \, 2^{\sqrt{x}} - (\log_2(x^2+2))(2^{\sqrt{x}})'}{(2^{\sqrt{x}})^2}$$

$$= \frac{\left(2x \cdot \frac{1}{(x^2+2)\ln 2}\right) 2^{\sqrt{x}} - (\log_2(x^2+2))(\ln 2 \cdot 2^{\sqrt{x}}) \cdot \frac{1}{2\sqrt{x}}}{2^{2\sqrt{x}}}$$

$$= \frac{2^{\sqrt{x}}\left(\frac{2x}{(x^2+2)\ln 2}\right) - 2^{\sqrt{x}} \cdot \ln 2 \cdot \frac{\log_2(x^2+2)}{2\sqrt{x}}}{2^{2\sqrt{x}}}$$

$$= \frac{\frac{2x}{(x^2+2)\ln 2} - \frac{\ln 2 \cdot \log_2(x^2+2)}{2\sqrt{x}}}{2^{\sqrt{x}}}$$

$$= \frac{2x}{2^{\sqrt{x}} \cdot \ln 2 (x^2+2)} - \frac{\ln 2 \cdot \log_2(x^2+2)}{2\sqrt{x} \cdot 2^{\sqrt{x}}}$$